

What is the central idea in Atomic structure?

Atomic number counts protons; nucleon number counts protons and neutrons. Isotopes have the same proton number but different neutron number.

9701-01-C1

QUESTION + ANSWER

State electron configuration of  $\text{Al}^{3+}$  ( $Z = 13$ ) and explain why its third ionisation energy is lower than its fourth.

$\text{Al}^{3+}$ :  $1s^2 2s^2 2p^6$ . First three electrons include valence  $3s^2 3p^1$ ; fourth is from inner  $n = 2$  shell and is held more strongly.

9701-01-P

QUESTION + ANSWER

0.20 mol  $\text{N}_2$  reacts with 0.45 mol  $\text{H}_2$ :  $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$ . Find limiting reagent and theoretical  $\text{NH}_3$  amount.

0.20 mol  $\text{N}_2$  needs 0.60 mol  $\text{H}_2$ , so  $\text{H}_2$  limits.  $\text{NH}_3 = (2/3)(0.45) = 0.30$  mol.

9701-02-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Chemical bonding?

VSEPR predicts electron-pair arrangement; lone pairs repel more strongly than bonding pairs. Hydrogen bonding requires H bonded to N, O or F and a suitable lone pair.

9701-03-C2

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Atomic structure?

Electron configurations fill orbitals according to energy, Pauli exclusion and Hund's rule. Successive ionisation energy jumps reveal an inner shell after removal of valence electrons.

9701-01-C2

QUESTION + ANSWER

What is the central idea in Atoms, molecules and stoichiometry?

Use  $n = m/M$ ,  $n = cV$  with  $V$  in  $\text{dm}^3$ , and gas molar volume only under specified conditions. Balance equations before mole ratios.

9701-02-C1

QUESTION + ANSWER

Which reagent limits when 1 mol  $\text{N}_2$  reacts with 1 mol  $\text{H}_2$  to make  $\text{NH}_3$ ?

- A Nitrogen
- B Hydrogen
- C Both equal
- D Neither

B. Hydrogen  
Why: Use  $n = m/M$ ,  $n = cV$  with  $V$  in  $\text{dm}^3$ , and gas molar volume only under specified conditions.

9701-02-M

QUESTION + ANSWER

Why is water polar?

- A It is linear
- B O-H bonds are nonpolar
- C Bent shape prevents dipole cancellation
- D It has no lone pairs

C. Bent shape prevents dipole cancellation  
Why: Ionic bonding is electrostatic attraction between oppositely charged ions.

9701-03-M

QUESTION + ANSWER

What is the electron configuration of  $\text{Mg}^{2+}$ ?

- A  $1s^2 2s^2 2p^6$
- B  $1s^2 2s^2 2p^6 3s^2$
- C  $1s^2 2s^2 2p^5$
- D  $1s^2 2s^2 2p^6 3s^1$

A.  $1s^2 2s^2 2p^6$   
Why: Atomic number counts protons; nucleon number counts protons and neutrons.

9701-01-M

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Atoms, molecules and stoichiometry?

A limiting reagent runs out first; compare available amount divided by stoichiometric coefficient. Percentage yield compares actual and theoretical product amounts.

9701-02-C2

QUESTION + ANSWER

What is the central idea in Chemical bonding?

Ionic bonding is electrostatic attraction between oppositely charged ions. Covalent bonds share electron pairs; electronegativity difference gives polarity, but molecular shape determines overall dipole.

9701-03-C1

QUESTION + ANSWER

Explain why water has a bent shape and a permanent dipole.

O has two bonding and two lone pairs; lone-pair repulsion produces bent arrangement; O-H bond dipoles do not cancel.

9701-03-P

QUESTION + ANSWER

What is the central idea in States of matter?

Ideal gas particles occupy negligible volume and have no intermolecular attraction.  $pV = nRT$  uses SI units and kelvin. Real gases deviate most at high pressure or low temperature where particle volume and attraction matter.

9701-04-C1

QUESTION + ANSWER

What is the central idea in Chemical energetics?

Exothermic reactions have negative enthalpy change; endothermic positive. Hess's law lets enthalpy changes be added because enthalpy is a state function.

9701-05-C1

QUESTION + ANSWER

A 0.050 mol reaction warms 200 g water by 5.0 K. Estimate molar  $\Delta H$  using  $c = 4.18 \text{ J g}^{-1} \text{ K}^{-1}$ .

$q = 200 \times 4.18 \times 5 = 4180 \text{ J}$ ;  $\Delta H = -4180/0.050 = -83\,600 \text{ J mol}^{-1} = -83.6 \text{ kJ mol}^{-1}$ .

9701-05-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Electrochemistry?

Standard electrode potentials describe reduction half-cells under standard conditions; positive  $E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$  indicates thermodynamic feasibility under standard conditions, not reaction rate.

9701-06-C2

QUESTION + ANSWER

Which mechanism, condition or distinction matters in States of matter?

Solids, liquids and gases differ in particle arrangement, movement and proximity, while intermolecular forces influence melting and boiling.

9701-04-C2

QUESTION + ANSWER

Calculate pressure of 0.050 mol ideal gas at 300 K in  $1.0 \text{ dm}^3$  ( $R = 8.31$ ).

$V = 0.0010 \text{ m}^3$ ;  $p = nRT/V = 1.25 \times 10^5 \text{ Pa}$ .

9701-04-P

QUESTION + ANSWER

Which sign represents an exothermic  $\Delta H$ ?

- A Negative
- B Positive always
- C Zero
- D Undefined

A. Negative  
Why: Exothermic reactions have negative enthalpy change; endothermic positive.

9701-05-M

QUESTION + ANSWER

Which statement identifies oxidation?

- A Gain of protons
- B Loss of electrons
- C Loss of neutrons
- D Gain of electrons

B. Loss of electrons  
Why: Oxidation is electron loss, reduction electron gain.

9701-06-M

QUESTION + ANSWER

Which temperature must be used in  $pV = nRT$ ?

- A Degrees Celsius
- B Degrees Fahrenheit
- C Any unconverted scale
- D Kelvin

D. Kelvin

Why: Ideal gas particles occupy negligible volume and have no intermolecular attraction.

9701-04-M

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Chemical energetics?

Bond energies approximate  $\Delta H = \text{bonds broken} - \text{bonds formed}$ ; a calorimeter uses  $q = mc\Delta T$ . Quote signs and account for losses when evaluating experimental values.

9701-05-C2

QUESTION + ANSWER

What is the central idea in Electrochemistry?

Oxidation is electron loss, reduction electron gain. Oxidation numbers help balance redox equations, but charge and atoms must also balance.

9701-06-C1

QUESTION + ANSWER

Given  $E^\circ(\text{Fe}^{3+}/\text{Fe}^{2+}) = +0.77 \text{ V}$  and  $E^\circ(\text{I}_2/\text{I}^-) = +0.54 \text{ V}$ , find  $E^\circ_{\text{cell}}$  and identify the species oxidised.

$\text{Fe}^{3+}$  reduced;  $\text{I}^-$  oxidised.  $E^\circ_{\text{cell}} = 0.77 - 0.54 = +0.23 \text{ V}$ .

9701-06-P

QUESTION + ANSWER

FagusED · A-Level 9701  
7 Equilibria

What is the central idea in Equilibria?

A dynamic equilibrium has equal forward and reverse rates in a closed system, though concentrations need not be equal.  $K_c$  uses equilibrium concentrations raised to stoichiometric powers; omit pure solids/liquids.

9701-07-C1

QUESTION + ANSWER

FagusED · A-Level 9701  
8 Reaction kinetics

What is the central idea in Reaction kinetics?

Rate can be measured from the tangent to concentration-time curves, gas collection or mass loss.

9701-08-C1

QUESTION + ANSWER

FagusED · A-Level 9701  
8 Reaction kinetics

Explain why a catalyst increases both forward and reverse reaction rates but does not change  $K_c$ .

It provides lower activation-energy route for both directions; equilibrium composition/thermodynamics at fixed T are unchanged.

9701-08-P

QUESTION + ANSWER

FagusED · A-Level 9701  
9 Periodicity

Which mechanism, condition or distinction matters in Periodicity?

Oxides change from basic/metallic through amphoteric to acidic/non-metallic. Explain anomalies with orbital energy and electron repulsion, not a blanket memorised trend.

9701-09-C2

QUESTION + ANSWER

FagusED · A-Level 9701  
10 Group 2

What is the central idea in Group 2?

Down Group 2, ionisation energy decreases due to increased distance and shielding. Metal reactivity with water increases.

9701-10-C1

QUESTION + ANSWER

FagusED · A-Level 9701  
7 Equilibria

Which mechanism, condition or distinction matters in Equilibria?

Changing temperature changes  $K_c$ ; changing concentration or pressure shifts composition while  $K_c$  stays fixed at fixed temperature.

9701-07-C2

QUESTION + ANSWER

FagusED · A-Level 9701  
7 Equilibria

For  $A + B \rightleftharpoons 2C$ , equilibrium concentrations are  $[A]=0.20$ ,  $[B]=0.50$ ,  $[C]=0.40 \text{ mol dm}^{-3}$ . Calculate  $K_c$ .

$K_c = [C]^2/([A][B]) = 0.40^2/(0.20 \times 0.50) = 1.6$ .

9701-07-P

QUESTION + ANSWER

FagusED · A-Level 9701  
8 Reaction kinetics

What is the effect of a catalyst on activation energy?

A Raises it  
B Eliminates collisions  
C Changes  $\Delta H$   
D Lowers it

D. Lowers it  
Why: Rate can be measured from the tangent to concentration-time curves, gas collection or mass loss.

9701-08-M

QUESTION + ANSWER

FagusED · A-Level 9701  
9 Periodicity

Which oxide is generally more acidic?

A  $SO_3$   
B  $Na_2O$   
C  $MgO$   
D  $Al_2O_3$  only

A.  $SO_3$   
Why: Across Period 3 nuclear charge increases and atomic radius generally decreases as shielding changes little; first ionisation energy broadly increases with subshell/pairing exceptions.

9701-09-M

QUESTION + ANSWER

FagusED · A-Level 9701  
7 Equilibria

What changes  $K_c$  for a fixed reaction?

A Adding more solid  
B Pressure at fixed T  
C Temperature  
D A catalyst at fixed T

C. Temperature  
Why: A dynamic equilibrium has equal forward and reverse rates in a closed system, though concentrations need not be equal.

9701-07-M

QUESTION + ANSWER

FagusED · A-Level 9701  
8 Reaction kinetics

Which mechanism, condition or distinction matters in Reaction kinetics?

Collision theory requires sufficient energy and suitable orientation; increasing temperature raises the fraction with energy above activation energy. Catalysts offer a lower activation energy pathway and do not alter equilibrium position or  $K_c$ .

9701-08-C2

QUESTION + ANSWER

FagusED · A-Level 9701  
9 Periodicity

What is the central idea in Periodicity?

Across Period 3 nuclear charge increases and atomic radius generally decreases as shielding changes little; first ionisation energy broadly increases with subshell/pairing exceptions.

9701-09-C1

QUESTION + ANSWER

FagusED · A-Level 9701  
9 Periodicity

Explain why first ionisation energy of S is lower than P despite greater nuclear charge.

In sulfur one 3p orbital has paired electrons; inter-electron repulsion makes one easier to remove than the unpaired 3p electron in phosphorus.

9701-09-P

QUESTION + ANSWER

FagusED · A-Level 9701  
10 Group 2

Which mechanism, condition or distinction matters in Group 2?

Hydroxide solubility rises down the group; sulfate solubility falls. Thermal decomposition of carbonates/nitrates becomes harder as larger cations polarise anions less strongly.

9701-10-C2

QUESTION + ANSWER

FagusED · A-Level 9701  
11 Group 17

What is the central idea in Group 17?

Halogen oxidising strength decreases down the group; halide reducing ability increases. A more reactive halogen displaces a less reactive halide from solution.

9701-11-C1

QUESTION + ANSWER

FagusED · A-Level 9701  
12 Nitrogen and sulfur

What is the central idea in Nitrogen and sulfur?

Ammonia's lone pair makes it a base and ligand. The Haber process balances equilibrium yield with rate and cost using moderate conditions and a catalyst; catalyst does not shift equilibrium.

9701-12-C1

QUESTION + ANSWER

FagusED · A-Level 9701  
12 Nitrogen and sulfur

Explain why industrial ammonia production uses a catalyst and recycles unreacted gases.

Catalyst speeds forward and reverse reactions; does not alter equilibrium yield; recycling unreacted  $N_2$  and  $H_2$  raises overall conversion.

9701-12-P

QUESTION + ANSWER

FagusED · A-Level 9701  
10 Group 2

How does Group 2 hydroxide solubility generally change down the group?

- A Decreases
- B Increases
- C Stays zero
- D Cannot be measured

B. Increases

Why: Down Group 2, ionisation energy decreases due to increased distance and shielding.

9701-10-M

QUESTION + ANSWER

FagusED · A-Level 9701  
11 Group 17

Which halogen oxidises  $Br^-$  to  $Br_2$  in aqueous solution?

- A  $I_2$
- B  $Br_2$
- C  $Cl_2$
- D  $Br^-$

C.  $Cl_2$

Why: Halogen oxidising strength decreases down the group; halide reducing ability increases.

9701-11-M

QUESTION + ANSWER

FagusED · A-Level 9701  
12 Nitrogen and sulfur

Which mechanism, condition or distinction matters in Nitrogen and sulfur?

Sulfur dioxide is an acidic, reducing oxide and an atmospheric pollutant; sulfuric acid manufacture involves reversible oxidation to  $SO_3$ .

9701-12-C2

QUESTION + ANSWER

FagusED · A-Level 9701  
13 Introduction to organic chemistry

What is the central idea in Introduction to organic chemistry?

Functional groups predict characteristic reactions. Structural isomers have the same molecular formula but different connectivity; stereoisomers have the same connectivity but different spatial arrangement.

9701-13-C1

QUESTION + ANSWER

FagusED · A-Level 9701  
10 Group 2

Explain why  $MgCO_3$  decomposes more readily than  $BaCO_3$  on heating.

$Mg^{2+}$  is smaller and more strongly polarises carbonate ion, weakening C-O bonding and lowering thermal stability.

9701-10-P

QUESTION + ANSWER

FagusED · A-Level 9701  
11 Group 17

Which mechanism, condition or distinction matters in Group 17?

Halide ions give silver halide precipitates with  $AgNO_3$  after acidification; colour and ammonia solubility help identify ions.

9701-11-C2

QUESTION + ANSWER

FagusED · A-Level 9701  
11 Group 17

Predict observation when chlorine water is added to aqueous potassium iodide, and write ionic equation.

Brown iodine appears;  $Cl_2 + 2I^- \rightarrow 2Cl^- + I_2$ .

9701-11-P

QUESTION + ANSWER

FagusED · A-Level 9701  
12 Nitrogen and sulfur

What does an industrial catalyst change in Haber equilibrium?

- A Position of equilibrium
- B Kc
- C Enthalpy change
- D Rate of attainment

D. Rate of attainment

Why: Ammonia's lone pair makes it a base and ligand.

9701-12-M

QUESTION + ANSWER

FagusED · A-Level 9701  
13 Introduction to organic chemistry

Which mechanism, condition or distinction matters in Introduction to organic chemistry?

Curly arrows show movement of an electron pair from its source to an electron-deficient site; distinguish electrophile, nucleophile and free radical.

9701-13-C2

QUESTION + ANSWER

FagusED · A-Level 9701  
13 Introduction to organic chemistry

Which pair can be E/Z isomers?  
A But-2-ene  
B Ethene  
C Propene  
D 2-methylpropene

A. But-2-ene  
Why: Functional groups predict characteristic reactions.

9701-13-M

QUESTION + ANSWER

FagusED · A-Level 9701  
14 Hydrocarbons

State the mechanism for adding HBr to propene and predict major product using carbocation stability.

Electrophilic addition;  $H^+$  adds to terminal carbon to form more stable secondary carbocation; major product 2-bromopropane.

9701-14-P

QUESTION + ANSWER

FagusED · A-Level 9701  
15 Halogen compounds

Which mechanism, condition or distinction matters in Halogen compounds?

Rate depends on bond strength and structure. Halogenoarenes resist substitution under mild conditions due to resonance affecting C-halogen bond.

9701-15-C2

QUESTION + ANSWER

FagusED · A-Level 9701  
16 Hydroxy compounds

What is the central idea in Hydroxy compounds?

Alcohol oxidation depends on carbon bearing  $-OH$ : primary gives aldehyde then acid; secondary gives ketone; tertiary resists mild oxidation.

9701-16-C1

QUESTION + ANSWER

FagusED · A-Level 9701  
13 Introduction to organic chemistry

Explain why  $CH_3CH=CHCH_3$  can have E/Z isomers but  $CH_2=CHCH_2CH_3$  cannot.

Each alkene carbon in but-2-ene has two different groups; terminal carbon of but-1-ene has two H atoms.

9701-13-P

QUESTION + ANSWER

FagusED · A-Level 9701  
14 Hydrocarbons

Which mechanism, condition or distinction matters in Hydrocarbons?

Benzene is stabilised by delocalised  $\pi$  electrons and typically substitutes rather than adds, preserving aromaticity.

9701-14-C2

QUESTION + ANSWER

FagusED · A-Level 9701  
15 Halogen compounds

What is the central idea in Halogen compounds?

Haloalkanes undergo nucleophilic substitution when a nucleophile attacks  $\delta^+$  carbon; aqueous hydroxide produces an alcohol, while ethanolic hydroxide favours elimination to alkene.

9701-15-C1

QUESTION + ANSWER

FagusED · A-Level 9701  
15 Halogen compounds

Predict organic product when 2-bromopropane reacts with hot ethanolic KOH and explain the role of base.

Propene;  $OH^-$  removes  $\beta$ -hydrogen, and HBr is eliminated to make  $C=C$ .

9701-15-P

QUESTION + ANSWER

FagusED · A-Level 9701  
14 Hydrocarbons

What is the central idea in Hydrocarbons?

Alkanes undergo free-radical substitution with halogens under UV light, often producing mixtures. Alkenes undergo electrophilic addition because  $\pi$  electrons attract electrophiles.

9701-14-C1

QUESTION + ANSWER

FagusED · A-Level 9701  
14 Hydrocarbons

Which reaction class describes HBr addition to propene?

A Free-radical substitution  
B Electrophilic addition  
C Elimination  
D Condensation

B. Electrophilic addition  
Why: Alkanes undergo free-radical substitution with halogens under UV light, often producing mixtures.

9701-14-M

QUESTION + ANSWER

FagusED · A-Level 9701  
15 Halogen compounds

Hot ethanolic  $OH^-$  typically promotes which haloalkane reaction?

A Oxidation  
B Polymerisation  
C Elimination  
D Ester hydrolysis

C. Elimination  
Why: Haloalkanes undergo nucleophilic substitution when a nucleophile attacks  $\delta^+$  carbon; aqueous hydroxide produces an alcohol, while ethanolic hydroxide favours elimination to alkene.

9701-15-M

QUESTION + ANSWER

FagusED · A-Level 9701  
16 Hydroxy compounds

Which mechanism, condition or distinction matters in Hydroxy compounds?

Conditions matter: distil aldehyde to prevent further oxidation, reflux for acid. Phenol is more acidic than an alcohol because phenoxide ion is resonance stabilised.

9701-16-C2

QUESTION + ANSWER

Which alcohol typically oxidises to a ketone?

- A Primary  
B Tertiary  
C Phenol only  
D Secondary

D. Secondary

Why: Alcohol oxidation depends on carbon bearing -OH: primary gives aldehyde then acid; secondary gives ketone; tertiary resists mild oxidation.

9701-16-M

QUESTION + ANSWER

An unknown  $C_3H_6O$  compound forms a silver mirror and is reduced to propan-1-ol. Identify it and justify.

Propanal; aldehyde reduces Tollens' reagent; carbonyl reduction gives primary alcohol.

9701-17-P

QUESTION + ANSWER

What is produced when ethanol reacts with ethanoic acid under suitable conditions?

- A Ethanamide  
B Ethyl ethanoate  
C Ethanoyl chloride  
D Ethene

B. Ethyl ethanoate

Why: Carboxylic acids donate  $H^+$  because carboxylate is resonance stabilised.

9701-18-M

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Nitrogen compounds?

Amines react with acids to form ammonium salts and with acyl chlorides to form amides. Amino acids have both amino and carboxyl groups and often exist as zwitterions depending on pH.

9701-19-C2

QUESTION + ANSWER

Outline conditions and product for oxidising ethanol to ethanoic acid rather than ethanal.

Use excess acidified dichromate(VI); heat under reflux; product ethanoic acid.

9701-16-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Carbonyl compounds?

$NaBH_4$  reduces aldehydes to primary alcohols and ketones to secondary alcohols. HCN addition creates cyanohydrins and may create a chiral centre.

9701-17-C2

QUESTION + ANSWER

What is the central idea in Carboxylic acids and derivatives?

Carboxylic acids donate  $H^+$  because carboxylate is resonance stabilised. Esters form by acid-catalysed condensation of acid and alcohol; hydrolysis reverses ester formation.

9701-18-C1

QUESTION + ANSWER

Name ester from propanoic acid and methanol, and state two conditions/features of preparation.

Methyl propanoate; acid catalyst e.g. concentrated  $H_2SO_4$ ; heat under reflux/reversible condensation with water.

9701-18-P

QUESTION + ANSWER

What is the central idea in Carbonyl compounds?

Aldehydes and ketones have polar  $C=O$  bonds susceptible to nucleophilic addition. Aldehydes oxidise with Tollens' or Fehling's reagents while ketones generally do not.

9701-17-C1

QUESTION + ANSWER

Which carbonyl compound forms a silver mirror with Tollens' reagent?

- A Aldehyde  
B Ketone only  
C Alkane  
D Ester only

A. Aldehyde

Why: Aldehydes and ketones have polar  $C=O$  bonds susceptible to nucleophilic addition.

9701-17-M

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Carboxylic acids and derivatives?

Acyl chlorides are more reactive than esters and undergo nucleophilic acyl substitution, giving HCl with alcohols or water.

9701-18-C2

QUESTION + ANSWER

What is the central idea in Nitrogen compounds?

Amines are bases because the nitrogen lone pair accepts  $H^+$ ; electron donation by alkyl groups can alter basicity.

9701-19-C1

QUESTION + ANSWER

**FagusED · A-Level 9701**  
**19 Nitrogen compounds**

What acts as a Brønsted base in amines?

- A N-H bond only
- B Carbon skeleton
- C Nitrogen lone pair
- D Carboxyl oxygen always

C. Nitrogen lone pair  
Why: Amines are bases because the nitrogen lone pair accepts  $H^+$ ; electron donation by alkyl groups can alter basicity.

9701-19-M

QUESTION + ANSWER

**FagusED · A-Level 9701**  
**20 Polymerisation**

Identify polymerisation type for ethane-1,2-diol and ethanedioic acid, and name the linkage and small molecule released.

Condensation; ester link; water.

9701-20-P

QUESTION + ANSWER

**FagusED · A-Level 9701**  
**21 Organic synthesis**

Which mechanism, condition or distinction matters in Organic synthesis?

Overall yield is the product of fractional step yields; purification may reduce isolated yield but increase purity. Check carbon count and oxidation state at every step.

9701-21-C2

QUESTION + ANSWER

**FagusED · A-Level 9701**  
**22 Analytical techniques**

What is the central idea in Analytical techniques?

Infrared absorption helps identify functional groups; a broad O-H band differs for alcohol and carboxylic acid.

9701-22-C1

QUESTION + ANSWER

**FagusED · A-Level 9701**  
**19 Nitrogen compounds**

Explain why glycine can migrate toward a cathode at sufficiently low pH and show its predominant charged form.

At low pH amino group is protonated and carboxyl group mainly  $COOH$ ;  $H_3N^+CH_2COOH$  has net + charge and moves toward cathode.

9701-19-P

QUESTION + ANSWER

**FagusED · A-Level 9701**  
**20 Polymerisation**

What small molecule is often lost during polyester formation?

- A  $O_2$
- B  $N_2$
- C  $Br_2$
- D  $H_2O$

D.  $H_2O$   
Why: Addition polymerisation joins alkene monomers without small-molecule by-products.

9701-20-M

QUESTION + ANSWER

**FagusED · A-Level 9701**  
**21 Organic synthesis**

How is overall yield of successive steps found?

- A Multiply fractional yields
- B Add percentage yields
- C Use only the first step
- D Divide yield by step count

A. Multiply fractional yields  
Why: A multi-step synthesis needs a feasible path between functional groups, with reagents and conditions chosen to control competing reactions.

9701-21-M

QUESTION + ANSWER

**FagusED · A-Level 9701**  
**22 Analytical techniques**

Which mechanism, condition or distinction matters in Analytical techniques?

Mass spectra provide molecular-ion mass and fragmentation clues; isotope patterns can indicate halogens.  $^1H$  NMR signals reflect chemically distinct proton environments, with integration showing relative counts and splitting from neighbouring protons.

9701-22-C2

QUESTION + ANSWER

**FagusED · A-Level 9701**  
**20 Polymerisation**

What is the central idea in Polymerisation?

Addition polymerisation joins alkene monomers without small-molecule by-products. Condensation polymerisation links bifunctional monomers and removes small molecules, commonly water or  $HCl$ .

9701-20-C1

QUESTION + ANSWER

**FagusED · A-Level 9701**  
**20 Polymerisation**

Which mechanism, condition or distinction matters in Polymerisation?

Hydrolysable ester and amide links make some condensation polymers biodegradable; properties depend on structure, crystallinity and cross-linking.

9701-20-C2

QUESTION + ANSWER

**FagusED · A-Level 9701**  
**21 Organic synthesis**

What is the central idea in Organic synthesis?

A multi-step synthesis needs a feasible path between functional groups, with reagents and conditions chosen to control competing reactions.

9701-21-C1

QUESTION + ANSWER

**FagusED · A-Level 9701**  
**21 Organic synthesis**

A synthesis has yields of 70%, 80% and 50% for three steps. Calculate overall percentage yield and give one reason isolated yield may be low.

$0.70 \times 0.80 \times 0.50 = 0.28$ , or 28%; side reactions, incomplete conversion or purification losses.

9701-21-P

QUESTION + ANSWER

FagusED · A-Level 9701  
22 Analytical techniques

Which IR feature suggests an O–H group?

- A Only a  $^{13}\text{C}$  NMR peak
- B Broad O–H absorption
- C A chlorine isotope pair
- D Absence of all absorption

B. Broad O–H absorption  
Why: Infrared absorption helps identify functional groups; a broad O–H band differs for alcohol and carboxylic acid.

9701-22-M

QUESTION + ANSWER

FagusED · A-Level 9701  
23 Chemical energetics

A process has  $\Delta H = +75 \text{ kJ mol}^{-1}$  and  $\Delta S = +0.150 \text{ kJ mol}^{-1} \text{ K}^{-1}$ . Find threshold temperature for  $\Delta G < 0$ .

Set  $\Delta G = 0$ ,  $T = \Delta H/\Delta S = 75/0.150 = 500 \text{ K}$ ; feasible by  $\Delta G$  criterion above 500 K.

9701-23-P

QUESTION + ANSWER

FagusED · A-Level 9701  
24 Electrochemistry

Which mechanism, condition or distinction matters in Electrochemistry?

For a spontaneous cell reaction  $\Delta G = -nFE$ , with  $n$  the electrons transferred. Electrolysis requires charge  $Q = It$  and amount electrons  $Q/F$ ; electron stoichiometry determines product amount.

9701-24-C2

QUESTION + ANSWER

FagusED · A-Level 9701  
25 Equilibria

What is the central idea in Equilibria?

Weak acid dissociation uses  $K_a = [\text{H}^+][\text{A}^-]/[\text{HA}]$ , while  $K_w = [\text{H}^+][\text{OH}^-]$ . A buffer contains a weak acid and conjugate base: added  $\text{H}^+$  is consumed by  $\text{A}^-$  and added  $\text{OH}^-$  by  $\text{HA}$ .

9701-25-C1

QUESTION + ANSWER

FagusED · A-Level 9701  
22 Analytical techniques

A compound shows molecular ion  $m/z$  46 and broad O–H IR absorption;  $^1\text{H}$  NMR has signals integrating 3:2:1. Suggest a structure and explain integration.

Ethanol  $\text{CH}_3\text{CH}_2\text{OH}$ ;  $\text{CH}_3$  gives 3 H,  $\text{CH}_2$  gives 2 H, OH gives 1 H.

9701-22-P

QUESTION + ANSWER

FagusED · A-Level 9701  
23 Chemical energetics

Which mechanism, condition or distinction matters in Chemical energetics?

A negative  $\Delta G$  does not ensure fast reaction. Standard enthalpy changes may be derived from formation data using products minus reactants.

9701-23-C2

QUESTION + ANSWER

FagusED · A-Level 9701  
24 Electrochemistry

What is the central idea in Electrochemistry?

The Nernst equation accounts for concentrations in electrode potential;  $E$  changes with reaction quotient rather than just the standard value.

9701-24-C1

QUESTION + ANSWER

FagusED · A-Level 9701  
24 Electrochemistry

Copper is deposited by  $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$  using 1.0 A for 1930 s. Find amount of Cu formed.

$Q = It = 1930 \text{ C}$ ; electrons =  $1930/96500 = 0.020 \text{ mol}$ ; Cu = 0.010 mol.

9701-24-P

QUESTION + ANSWER

FagusED · A-Level 9701  
23 Chemical energetics

What is the central idea in Chemical energetics?

At A Level, entropy measures dispersal of energy and arrangements;  $\Delta G = \Delta H - T\Delta S$  predicts feasible direction at specified temperature under standard conditions.

9701-23-C1

QUESTION + ANSWER

FagusED · A-Level 9701  
23 Chemical energetics

What criterion of  $\Delta G$  supports feasibility at specified conditions?

- A  $\Delta G > 0$  always
- B  $\Delta G = \Delta H$  only
- C  $\Delta G < 0$
- D  $\Delta G$  depends only on catalyst

C.  $\Delta G < 0$   
Why: At A Level, entropy measures dispersal of energy and arrangements;  $\Delta G = \Delta H - T\Delta S$  predicts feasible direction at specified temperature under standard conditions.

9701-23-M

QUESTION + ANSWER

FagusED · A-Level 9701  
24 Electrochemistry

How many moles electrons deposit 1 mol Cu from  $\text{Cu}^{2+}$ ?

- A 0.5
- B 1
- C 3
- D 2

D. 2  
Why: The Nernst equation accounts for concentrations in electrode potential;  $E$  changes with reaction quotient rather than just the standard value.

9701-24-M

QUESTION + ANSWER

FagusED · A-Level 9701  
25 Equilibria

Which mechanism, condition or distinction matters in Equilibria?

For acid buffer,  $\text{pH} = \text{pK}_a + \log([\text{A}^-]/[\text{HA}])$  when conditions justify approximation.

9701-25-C2

QUESTION + ANSWER

For equal  $[HA]$  and  $[A^-]$  in an acid buffer, pH approximately equals what?

- A  $pK_a$   
 B  $K_w$   
 C 0  
 D 14 always

A.  $pK_a$

Why: Weak acid dissociation uses  $K_a = \frac{[H^+][A^-]}{[HA]}$ , while  $K_w = \frac{[H^+][OH^-]}{[H_2O]}$ .

9701-25-M

QUESTION + ANSWER

When  $[A]$  doubles at fixed  $[B]$ , rate rises fourfold; when  $[B]$  doubles, rate is unchanged. Write rate law and overall order.

Order 2 in A, order 0 in B; rate =  $k[A]^2$ , overall order 2.

9701-26-P

QUESTION + ANSWER

Which compound forms a white insoluble precipitate in the sulfate test?

- A  $Mg(OH)_2$   
 B  $CaO$   
 C  $BaSO_4$   
 D  $Ba(OH)_2$

C.  $BaSO_4$

Why: For Group 2 compounds, ionic size and charge density explain solubility and thermal stability trends; hydration and lattice enthalpies compete in dissolving salts.

9701-27-M

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Transition elements?

Ligands donate electron pairs to a central ion; ligand substitution can change colour and coordination geometry. Colour depends on splitting and absorption of visible light;  $d^0/d^{10}$  ions are often colourless.

9701-28-C2

QUESTION + ANSWER

An acid buffer has  $pK_a = 4.76$  and  $[A^-]/[HA] = 10$ . Determine pH and explain effect of adding a little HCl.

$pH = 4.76 + \log_{10}(10) = 5.76$ ;  $A^-$  consumes much of added  $H^+$ , reducing pH change.

9701-25-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Reaction kinetics?

A first-order reaction has constant half-life at fixed temperature. The Arrhenius equation links temperature, activation energy and rate constant; a catalyst lowers activation energy.

9701-26-C2

QUESTION + ANSWER

What is the central idea in Group 2?

For Group 2 compounds, ionic size and charge density explain solubility and thermal stability trends; hydration and lattice enthalpies compete in dissolving salts.

9701-27-C1

QUESTION + ANSWER

Predict solubility trend of  $Mg(OH)_2$  to  $Ba(OH)_2$  and explain why sulfate tests use barium salts.

Hydroxide solubility increases down Group 2;  $BaSO_4$  has very low solubility, forming a visible white precipitate with sulfate ions.

9701-27-P

QUESTION + ANSWER

What is the central idea in Reaction kinetics?

A rate law is determined experimentally: rate =  $k[A]^m[B]^n$ . Reaction order is not inferred automatically from balanced stoichiometry.

9701-26-C1

QUESTION + ANSWER

When doubling  $[A]$  quadruples rate, order in A is what?

- A First  
 B Second  
 C Zero  
 D Fourth

B. Second

Why: A rate law is determined experimentally: rate =  $k[A]^m[B]^n$ .

9701-26-M

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Group 2?

Group 2 hydroxides become more soluble down the group; sulfate solubility decreases. Industrial/analytical uses depend on insolubility and acid-base behaviour.

9701-27-C2

QUESTION + ANSWER

What is the central idea in Transition elements?

Transition elements have atoms or stable ions with incomplete d subshells. Variable oxidation states, coloured complexes and catalysis arise from d-electron behaviour.

9701-28-C1

QUESTION + ANSWER

Which ion has incomplete d subshell?

- A  $\text{Zn}^{2+}$   
 B  $\text{Sc}^{3+}$   
 C  $\text{Cu}^+$   
 D  $\text{Cu}^{2+}$

D.  $\text{Cu}^{2+}$

Why: Transition elements have atoms or stable ions with incomplete d subshells.

9701-28-M

QUESTION + ANSWER

Why is a racemic mixture optically inactive even if each pure enantiomer rotates plane-polarised light?

The mixture has equal amounts of opposite enantiomers; rotations are equal and opposite so cancel.

9701-29-P

QUESTION + ANSWER

What is the electrophile in benzene nitration?

- A  $\text{NO}_3^-$   
 B  $\text{NO}_2^+$   
 C  $\text{NH}_4^+$   
 D  $\text{OH}^-$

B.  $\text{NO}_2^+$

Why: Aromatic reactions commonly proceed by electrophilic substitution to preserve benzene's delocalised  $\pi$  system.

9701-30-M

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Halogen compounds?

Haloalkane elimination competes with substitution; solvent, temperature and substrate structure influence outcome.

9701-31-C2

QUESTION + ANSWER

Explain why  $\text{Zn}^{2+}$  is usually colourless and  $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$  coloured.

$\text{Zn}^{2+}$  has full  $3d^{10}$  so no d-d transitions;  $\text{Cu}^{2+}$  has incomplete 3d subshell; ligand-field splitting allows visible photon absorption.

9701-28-P

QUESTION + ANSWER

Why is a racemate optically inactive?

- A Equal opposite rotations cancel  
 B No chiral molecules exist  
 C Molecules are all planar  
 D It cannot absorb light

A. Equal opposite rotations cancel  
 Why: At A Level, organic pathways integrate reaction mechanisms, stereochemistry and aromaticity.

9701-29-M

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Hydrocarbons?

Nitration uses nitronium ion generated by concentrated nitric and sulfuric acids; Friedel-Crafts reactions use a catalyst. Electron-donating/withdrawing substituents alter ring reactivity and direct substitution position.

9701-30-C2

QUESTION + ANSWER

What is the central idea in Halogen compounds?

Aryl halides are less reactive in nucleophilic substitution than haloalkanes because carbon-halogen bonding to the aromatic ring is strengthened by resonance and the  $\text{sp}^2$  carbon environment.

9701-31-C1

QUESTION + ANSWER

What is the central idea in Introduction to organic chemistry?

At A Level, organic pathways integrate reaction mechanisms, stereochemistry and aromaticity.

9701-29-C1

QUESTION + ANSWER

Which mechanism, condition or distinction matters in Introduction to organic chemistry?

Optical isomerism occurs when a tetrahedral carbon has four different groups; a racemate has equal enantiomers and no net optical rotation. Mechanisms must account for electron movement and the stability of intermediates.

9701-29-C2

QUESTION + ANSWER

What is the central idea in Hydrocarbons?

Aromatic reactions commonly proceed by electrophilic substitution to preserve benzene's delocalised  $\pi$  system.

9701-30-C1

QUESTION + ANSWER

Name electrophile in benzene nitration and explain why substitution predominates over addition.

$\text{NO}_2^+$ ; addition would destroy the stable delocalised aromatic system, while substitution restores it after deprotonation.

9701-30-P

QUESTION + ANSWER

FagusED · A-Level 9701  
31 Halogen compounds

Which conditions promote haloalkane elimination?

- A Cold aqueous OH<sup>-</sup>
- B Aqueous AgNO<sub>3</sub>
- C Hot ethanolic OH<sup>-</sup>
- D Dry HCl only

C. Hot ethanolic OH<sup>-</sup>  
Why: Aryl halides are less reactive in nucleophilic substitution than haloalkanes because carbon-halogen bonding to the aromatic ring is strengthened by resonance and the sp<sup>2</sup> carbon environment.

9701-31-M QUESTION + ANSWER

FagusED · A-Level 9701  
32 Hydroxy compounds

Phenol and ethanol both react with Na metal. Give one reagent that distinguishes them and expected result.

Aqueous NaOH; phenol dissolves/reacts forming phenoxide; ethanol shows no comparable acid-base reaction.

9701-32-P QUESTION + ANSWER

FagusED · A-Level 9701  
33 Carboxylic acids and derivatives

What is a by-product of ethanoyl chloride reacting with ethanol?

- A HCl
- B H<sub>2</sub>
- C O<sub>2</sub>
- D NH<sub>3</sub>

A. HCl  
Why: Acyl chlorides, acid anhydrides, esters and amides vary in ease of nucleophilic acyl substitution.

9701-33-M QUESTION + ANSWER

FagusED · A-Level 9701  
34 Nitrogen compounds

Which mechanism, condition or distinction matters in Nitrogen compounds?

Amino acids exist as zwitterions near their isoelectric point, affecting solubility and electrophoretic movement. Diazonium compounds can form azo dyes via coupling with activated aromatic rings under suitable cold conditions.

9701-34-C2 QUESTION + ANSWER

FagusED · A-Level 9701  
31 Halogen compounds

Describe two changes to favour elimination rather than substitution of 2-bromobutane and give two possible alkene products.

Use ethanolic rather than aqueous base and heat; but-1-ene and but-2-ene.

9701-31-P QUESTION + ANSWER

FagusED · A-Level 9701  
32 Hydroxy compounds

Which reagent reacts with phenol but typically not ethanol in acid-base test?

- A Water
- B Sodium metal
- C Bromine water only
- D Aqueous NaOH

D. Aqueous NaOH  
Why: Phenol is more acidic than aliphatic alcohol because the conjugate phenoxide ion delocalises negative charge.

9701-32-M QUESTION + ANSWER

FagusED · A-Level 9701  
33 Carboxylic acids and derivatives

Which mechanism, condition or distinction matters in Carboxylic acids and derivatives?

Amides are resonance stabilised and less reactive; hydrolysis may require acid/base and heat. Amino acids condense to peptide bonds.

9701-33-C2 QUESTION + ANSWER

FagusED · A-Level 9701  
34 Nitrogen compounds

What is the central idea in Nitrogen compounds?

Amines act as bases and nucleophiles; amides are much less basic because the nitrogen lone pair delocalises into the carbonyl.

9701-34-C1 QUESTION + ANSWER

FagusED · A-Level 9701  
32 Hydroxy compounds

What is the central idea in Hydroxy compounds?

Phenol is more acidic than aliphatic alcohol because the conjugate phenoxide ion delocalises negative charge. Phenol activates its ring toward bromination and nitration.

9701-32-C1 QUESTION + ANSWER

FagusED · A-Level 9701  
32 Hydroxy compounds

Which mechanism, condition or distinction matters in Hydroxy compounds?

Comparing reactions with Na, NaOH and Na<sub>2</sub>CO<sub>3</sub> distinguishes acidity levels: typical phenol reacts with NaOH but not sodium carbonate under usual conditions.

9701-32-C2 QUESTION + ANSWER

FagusED · A-Level 9701  
33 Carboxylic acids and derivatives

What is the central idea in Carboxylic acids and derivatives?

Acyl chlorides, acid anhydrides, esters and amides vary in ease of nucleophilic acyl substitution. Nucleophile attacks carbonyl C, tetrahedral intermediate forms and leaving group departs.

9701-33-C1 QUESTION + ANSWER

FagusED · A-Level 9701  
33 Carboxylic acids and derivatives

Name product from ethanoyl chloride and ethanol and describe the reaction type and another product.

Ethyl ethanoate; nucleophilic acyl substitution/esterification; HCl.

9701-33-P QUESTION + ANSWER

**FagusED · A-Level 9701**  
**34 Nitrogen compounds**

Which is more basic in water?

- A Ethanamide
- B Ethanamine
- C Both equal by definition
- D Neither can protonate

B. Ethanamine

Why: Amines act as bases and nucleophiles; amides are much less basic because the nitrogen lone pair delocalises into the carbonyl.

9701-34-M

QUESTION + ANSWER

**FagusED · A-Level 9701**  
**35 Polymerisation**

Explain why a polyamide may be hydrolysed by aqueous acid but poly(ethene) is relatively resistant.

Amide C-N links undergo hydrolysis; acid catalyses cleavage; poly(ethene) C-C backbone lacks such hydrolysable functional groups.

9701-35-P

QUESTION + ANSWER

**FagusED · A-Level 9701**  
**36 Organic synthesis**

Which mechanism, condition or distinction matters in Organic synthesis?

Forward route needs reagents, solvents and conditions; include isolation and purification. Percentage yields compound across steps and can make short routes preferable even if individual reactions look efficient.

9701-36-C2

QUESTION + ANSWER

**FagusED · A-Level 9701**  
**37 Analytical techniques**

What is the central idea in Analytical techniques?

A Level spectra may combine high-resolution mass spectrometry, IR,  $^1\text{H}/^{13}\text{C}$  NMR and chromatography. Use molecular formula/degree of unsaturation first, then identify carbonyl and O-H signals and proton environments; no single peak proves an entire structure.

9701-37-C1

QUESTION + ANSWER

**FagusED · A-Level 9701**  
**34 Nitrogen compounds**

Compare basicity of ethanamine and ethanamide using their nitrogen lone pairs.

Ethanamine is more basic; its N lone pair is available to accept  $\text{H}^+$ ; amide N lone pair delocalises into  $\text{C}=\text{O}$  and is less available.

9701-34-P

QUESTION + ANSWER

**FagusED · A-Level 9701**  
**35 Polymerisation**

Which polymer backbone lacks hydrolysable ester or amide links?

- A Polyester
- B Polyamide
- C Poly(ethene)
- D Protein

C. Poly(ethene)

Why: Condensation polyamides and polyesters are vulnerable to hydrolysis because their links can be cleaved; addition polymers with C-C backbones resist hydrolysis.

9701-35-M

QUESTION + ANSWER

**FagusED · A-Level 9701**  
**36 Organic synthesis**

What is a necessary check in retrosynthetic route planning?

- A Ignore carbon count
- B Only estimate colour
- C Never select conditions
- D Account for functional-group changes

D. Account for functional-group changes

Why: Plan retrosynthetically from the target, identify the key bond/functional group, then reverse to feasible starting materials.

9701-36-M

QUESTION + ANSWER

**FagusED · A-Level 9701**  
**37 Analytical techniques**

Which mechanism, condition or distinction matters in Analytical techniques?

Chromatographic  $R_f$  = distance spot/distance solvent front for the stated stationary and mobile phases.

9701-37-C2

QUESTION + ANSWER

**FagusED · A-Level 9701**  
**35 Polymerisation**

What is the central idea in Polymerisation?

Condensation polyamides and polyesters are vulnerable to hydrolysis because their links can be cleaved; addition polymers with C-C backbones resist hydrolysis.

9701-35-C1

QUESTION + ANSWER

**FagusED · A-Level 9701**  
**35 Polymerisation**

Which mechanism, condition or distinction matters in Polymerisation?

A repeat unit should show bonds that extend to adjacent units, and monomers can be deduced by cutting linkage bonds and restoring their functional groups.

9701-35-C2

QUESTION + ANSWER

**FagusED · A-Level 9701**  
**36 Organic synthesis**

What is the central idea in Organic synthesis?

Plan retrosynthetically from the target, identify the key bond/functional group, then reverse to feasible starting materials.

9701-36-C1

QUESTION + ANSWER

**FagusED · A-Level 9701**  
**36 Organic synthesis**

A three-step synthesis gives 80% yield at each step. Calculate overall yield and compare with a one-step 50% yield.

$0.8^3 = 0.512$  or 51.2%; three-step route has slightly greater overall yield than 50%.

9701-36-P

QUESTION + ANSWER

An M and M+2 mass peak in near 3:1 ratio suggests which atom?

- A Chlorine
- B Carbon only
- C Nitrogen
- D Hydrogen

A. Chlorine

Why: A Level spectra may combine high-resolution mass spectrometry, IR,  $^1\text{H}/^{13}\text{C}$  NMR and chromatography.

9701-37-M

QUESTION + ANSWER

A mean  $25.0\text{ cm}^3$  titre of  $0.100\text{ mol dm}^{-3}$  HCl neutralises  $25.0\text{ cm}^3$  NaOH. Determine NaOH concentration and state one titration good practice.

$n(\text{HCl}) = 0.00250\text{ mol}$ ; 1:1 gives  $[\text{NaOH}] = 0.100\text{ mol dm}^{-3}$ ; use concordant titres/read burette to two decimal places.

9701-38-P

QUESTION + ANSWER

For a fair rate comparison, what must be controlled?

- A Two variables changed together
- B Initial-rate units omitted
- C Temperature
- D Graph slope hidden

C. Temperature

Why: A plan identifies reaction, independent and dependent variables, controls and a measurable endpoint.

9701-39-M

QUESTION + ANSWER

A mass spectrum suggests formula  $\text{C}_3\text{H}_6\text{O}$  and IR has strong  $\text{C}=\text{O}$ ;  $^1\text{H}$  NMR has one proton environment. Suggest structure and justify.

Propanone  $\text{CH}_3\text{COCH}_3$ ; equivalent methyl groups give one  $^1\text{H}$  environment; carbonyl accounts for IR and formula.

9701-37-P

QUESTION + ANSWER

Which mechanism, condition or distinction matters in P3 Advanced practical skills?

Qualitative tests need reagent, observation and inference, plus alternatives where observations overlap.

9701-38-C2

QUESTION + ANSWER

What is the central idea in P5 Planning, analysis and evaluation?

A plan identifies reaction, independent and dependent variables, controls and a measurable endpoint.

9701-39-C1

QUESTION + ANSWER

Outline a fair comparison of reaction rate at two acid concentrations using gas volume.

Keep acid volume, solid mass/area and temperature constant; change acid concentration; measure initial gradient of gas-volume against time with repeat trials.

9701-39-P

QUESTION + ANSWER

What is the central idea in P3 Advanced practical skills?

Use safe measured quantities and observations to make reproducible chemical inferences. In titration rinse burette with titrant, pipette with analyte and repeat to concordance; record readings with appropriate decimal places.

9701-38-C1

QUESTION + ANSWER

Which good titration practice improves reliability?

- A One rough titre only
- B Repeat to concordance
- C Read volume to nearest litre
- D Rinse burette with analyte

B. Repeat to concordance

Why: Use safe measured quantities and observations to make reproducible chemical inferences.

9701-38-M

QUESTION + ANSWER

Which mechanism, condition or distinction matters in P5 Planning, analysis and evaluation?

Use a graph to distinguish orders or calculate an enthalpy with  $q = mc\Delta T$ , then evaluate heat losses and significant uncertainty. Safety controls should address actual reagents, not an unconnected generic warning.

9701-39-C2

QUESTION + ANSWER